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UPPER AUSTRIA

# MC527 Wearable Computing

## Activity Classification

HAGENBERG | LINZ | STEYR | WELS

# Agenda

- Motivation
- Machine learning with WEKA and scikit-learn
  - Introduction
  - Classification
  - Regression
- Activity recognition using cell phone accelerometers

# Sitting, physical activity and mortality

- U. Ekelund et. al. Does physical activity attenuate, or even eliminate, the detrimental association of sitting time with mortality? A harmonised meta-analysis of data from more than 1 million men and women. Lancet 2016.

- Does being physically active change the association between prolonged sitting and risk of death?

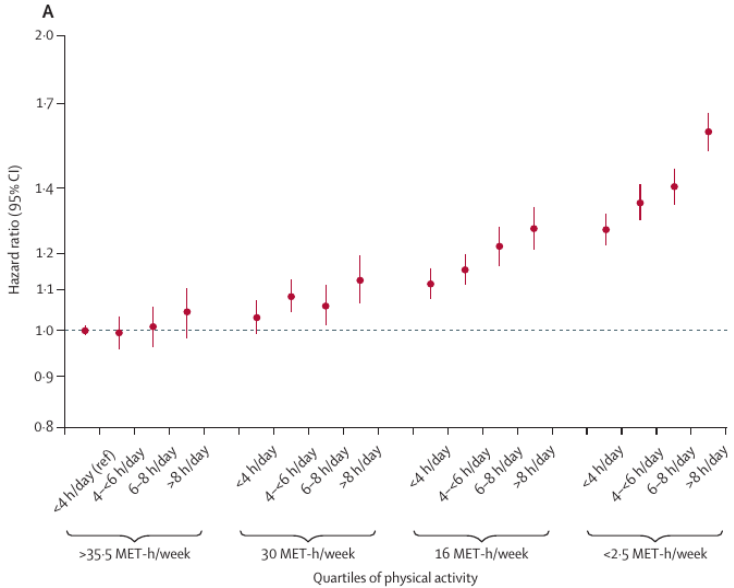
- Main findings
  - High activity attenuated the association with sitting (daily activity about 60–75 min per day)
  - Heavy TV viewing still showed excess mortality (daily watching time > 3h)

# Sitting, physical activity and mortality

- Study design and scope
  - 16 prospective cohorts (prospective – following people forward in time, recording outcomes after measuring their initial behaviors or characteristics)
  - Sitting analysis
    - 13 cohorts, 1,005,791 participants
    - 84,609 deaths
    - Follow-up: 2–18.1 years
  - TV-viewing analysis
    - 6, cohorts, 465,450 participants
    - 43,740 deaths
    - Follow-up: 5.8–13.7 years

# Sitting, physical activity and mortality

- High activity attenuated the sitting association



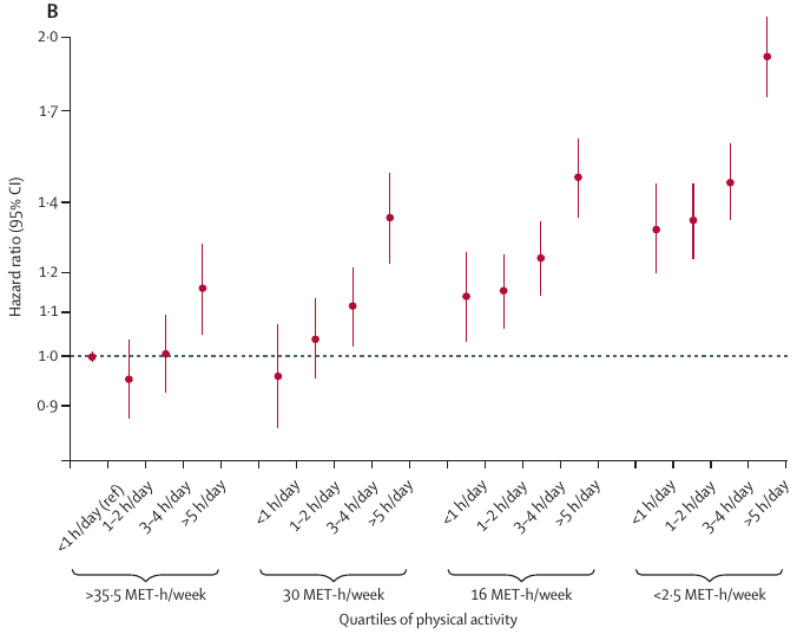

Most active: ≈60–75 min/day



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# Sitting, physical activity and mortality

- Heavy TV viewing retained an association




Even in the most active group:
≥5 h/day watching TV
HR 1.16 -> 16% higher instantaneous death rate



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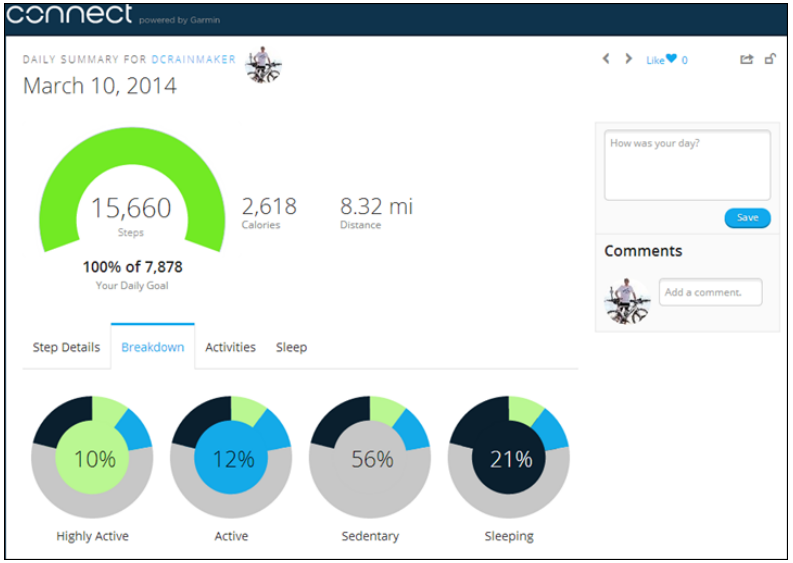
# Sitting, physical activity and mortality

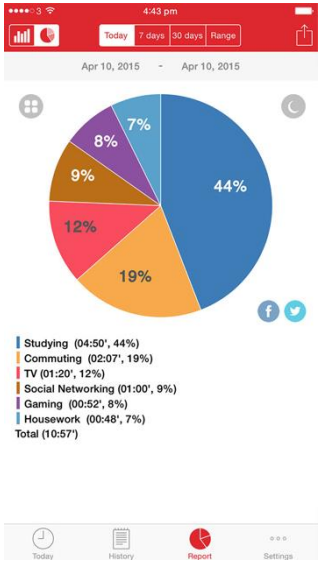
- Author's explanations, not tested in this meta-analysis
  - **After meals**. Evening TV often follows dinner. Prolonged sitting then may affect glucose and fat metabolism.
  - **Fewer breaks**. People may interrupt sitting more often at work than while watching TV.
  - **Eating habits**. Snacking and food advertising may contribute to the stronger association.
  - **Reporting accuracy.** Differences in reporting TV time and total sitting may partly explain the different estimates.

# Motivation

- Activity trackers could help their wearers to cut down on sitting time



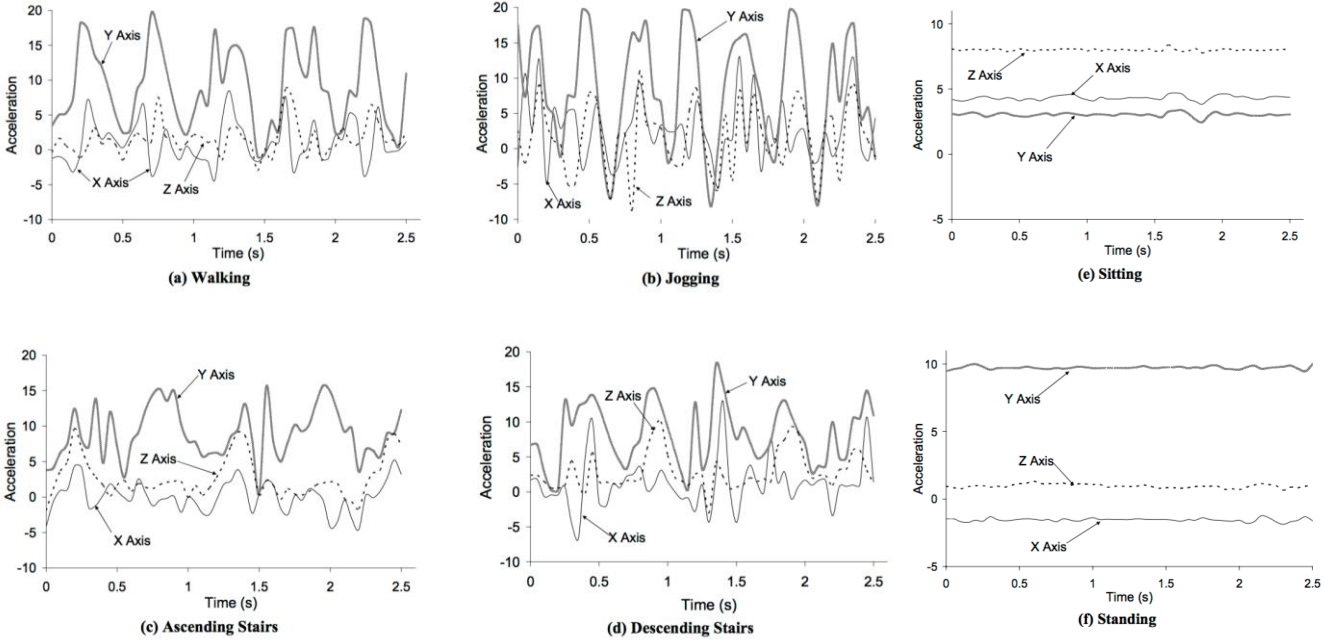






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# Motivation

- Accelerometer data for various activities





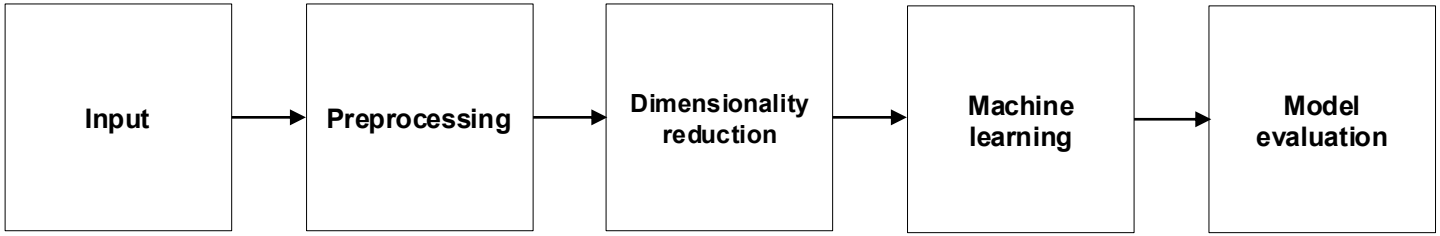

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# Motivation

- How to differentiate between the activities? How to code the rules?
  - Manually
    - You can determine a target value by using simple rules, computations, or predetermined steps that can be programmed without needing any data-driven learning
  - Machine learning techniques (ML)
    - You cannot code the rules: many factors could influence the answer. Rules depend on too many factors and many of these rules overlap or need to be tuned very finely
    - You cannot scale: You might be able to manually recognize a few hundred emails and decide whether they are spam or not. However, this task becomes tedious for millions of emails.

# Machine Learning

- Big picture






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# WEKA

- **W**aikato **E**nvironment for **K**nowledge **A**nalysis, a software workbench which incorporates several standard machine learning techniques developed at the University of Waikato
  - GUI – nice for learning machine learning
  - Java-based – you can use the developed models in your Java applications

- https://waikato.github.io/weka-wiki/downloading_weka/

# Scikit-learn

- Machine Learning in Python
  - Programming skills required
  - Used a lot in real life

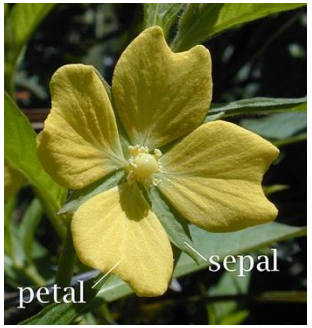
- https://scikit-learn.org/



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# Iris Dataset

- How to distinguish the species from each other?
  - Iris setosa, Iris versicolor, and Iris virginica
  - Based on the length and width of sepals and petals






Iris setosa



Iris versicolor

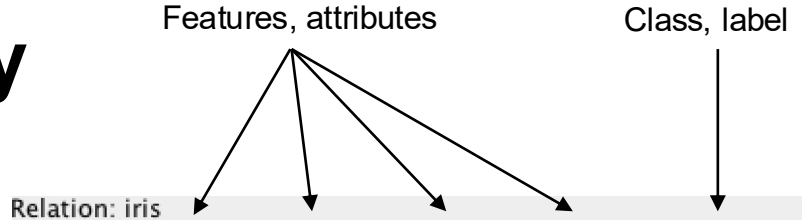


Iris virginica



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# Terminology




Instance, feature vector

Relation: iris

| No. | sepallength Numeric | sepalwidth Numeric | petallength Numeric | petalwidth Numeric | class Nominal |
|---|---|---|---|---|---|
| 1 | 5.1 | 3.5 | 1.4 | 0.2 | Iris-setosa |
| 2 | 4.9 | 3.0 | 1.4 | 0.2 | Iris-setosa |
| 3 | 4.7 | 3.2 | 1.3 | 0.2 | Iris-setosa |
| 4 | 4.6 | 3.1 | 1.5 | 0.2 | Iris-setosa |
| 5 | 5.0 | 3.6 | 1.4 | 0.2 | Iris-setosa |
| 6 | 5.4 | 3.9 | 1.7 | 0.4 | Iris-setosa |
| 7 | 4.6 | 3.4 | 1.4 | 0.3 | Iris-setosa |
| 8 | 5.0 | 3.4 | 1.5 | 0.2 | Iris-setosa |
| 9 | 4.4 | 2.9 | 1.4 | 0.2 | Iris-setosa |
| 10 | 4.9 | 3.1 | 1.5 | 0.1 | Iris-setosa |
| 11 | 5.4 | 3.7 | 1.5 | 0.2 | Iris-setosa |
| 12 | 4.8 | 3.4 | 1.6 | 0.2 | Iris-setosa |
| 13 | 4.8 | 3.0 | 1.4 | 0.1 | Iris-setosa |
| 14 | 4.3 | 3.0 | 1.1 | 0.1 | Iris-setosa |
| 15 | 5.8 | 4.0 | 1.2 | 0.2 | Iris-setosa |
| 16 | 5.7 | 4.4 | 1.5 | 0.4 | Iris-setosa |

Training dataset

Testing dataset



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# ARFF file

- Weka uses ARFF files (Attribute-Relation File Format)

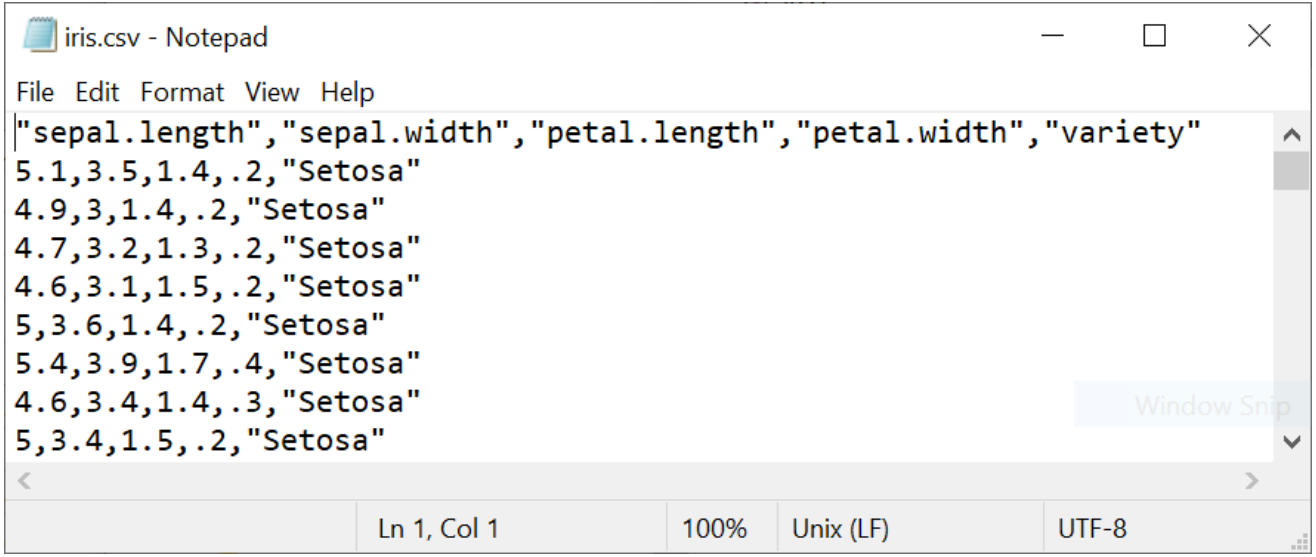
```
@RELATION iris
@ATTRIBUTE sepallength  REAL
@ATTRIBUTE sepalwidth   REAL
@ATTRIBUTE petallength  REAL
@ATTRIBUTE petalwidth   REAL
@ATTRIBUTE class {Iris-setosa,Iris-versicolor,Iris-virginica}
@DATA
5.1,3.5,1.4,0.2,Iris-setosa
4.9,3.0,1.4,0.2,Iris-setosa
4.7,3.2,1.3,0.2,Iris-setosa
4.6,3.1,1.5,0.2,Iris-setosa
5.0,3.6,1.4,0.2,Iris-setosa
5.4,3.9,1.7,0.4,Iris-setosa
```



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# Reading CSV files in scikit-learn

- In scikit-learn, csv files can be used






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# Reading CSV files in scikit-learn

- You'll probably use pandas to handle tabular data
  - DataFrame, 2D-data with labels

```
import pandas as pd

iris = pd.read_csv('./iris.csv')

print(iris)
```

```
     sepal.length  sepal.width  petal.length  petal.width     variety
0             5.1          3.5           1.4          0.2      Setosa
1             4.9          3.0           1.4          0.2      Setosa
2             4.7          3.2           1.3          0.2      Setosa
3             4.6          3.1           1.5          0.2      Setosa
4             5.0          3.6           1.4          0.2      Setosa
..            ...          ...           ...          ...         ...
145           6.7          3.0           5.2          2.3   Virginica
146           6.3          2.5           5.0          1.9   Virginica
147           6.5          3.0           5.2          2.0   Virginica
148           6.2          3.4           5.4          2.3   Virginica
149           5.9          3.0           5.1          1.8   Virginica
```



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# Reading CSV files in scikit-learn

- You'll probably use pandas to handle tabular data
  - DataFrame, 2D-data with labels

```
import pandas as pd

iris = pd.read_csv('./iris.csv')

print(iris)

print(iris['sepal.length'])
```

```
[150 rows x 5 columns]
0      5.1
1      4.9
2      4.7
3      4.6
4      5.0
      ...
145    6.7
146    6.3
147    6.5
148    6.2
149    5.9
Name: sepal.length, Length: 150, dtype: float64
```



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# Supervised learning

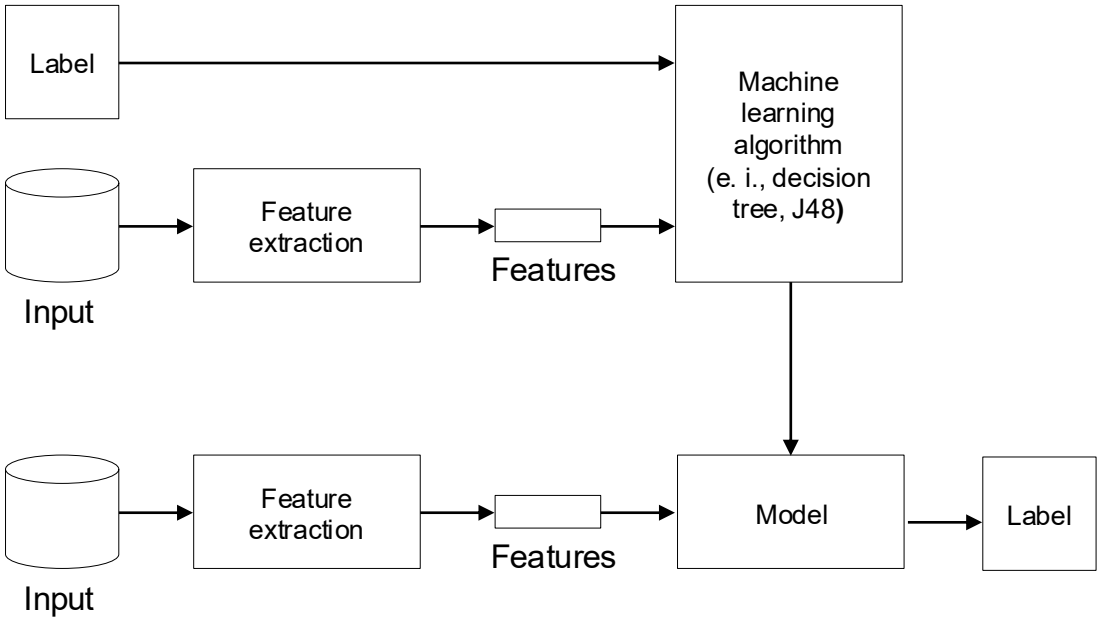
- Goal: make predictions for a new sample described by its attributes
  - Symbolic output: classification
  - Numerical output: regression



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# Supervised learning

- Classification




**Input**
5.1,3.5,1.4,0.2,Iris-setosa
4.9,3.0,1.4,0.2,Iris-setosa
4.7,3.2,1.3,0.2,Iris-setosa
4.6,3.1,1.5,0.2,Iris-setosa
5.0,3.6,1.4,0.2,Iris-setosa

**Output**
Model, algorithm

**Input**
5.0,3.4,4.0,0.3, ????

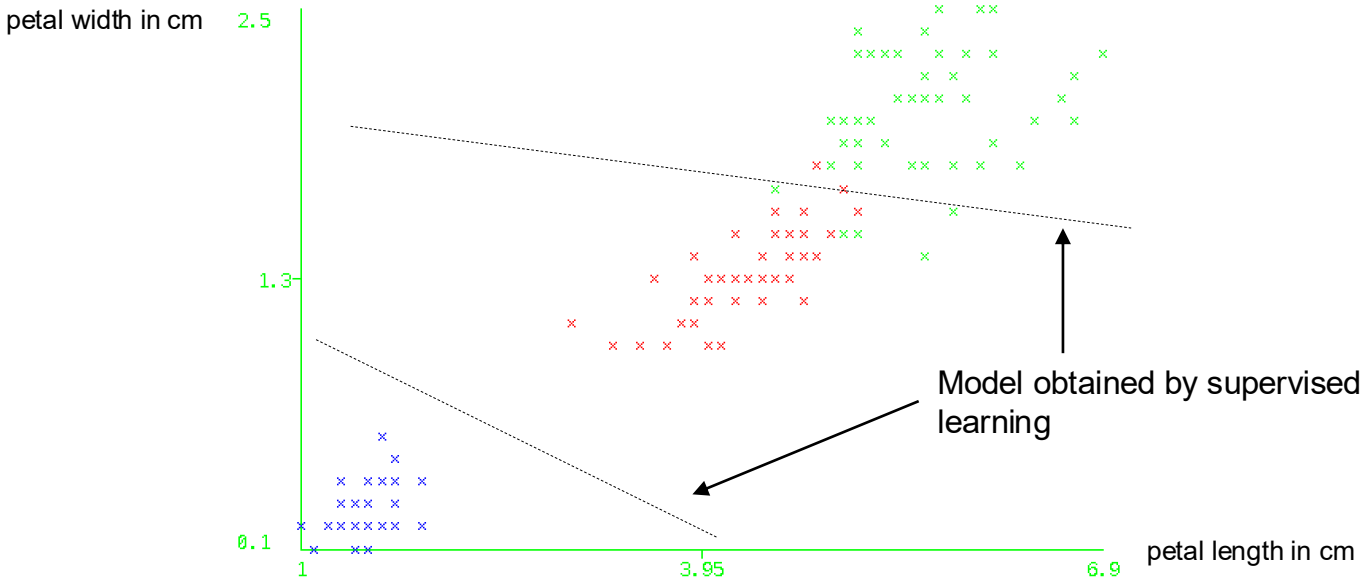
**Output**
Iris-setosa



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# Supervised learning

- Classification: predict class (e.g., Iris-setosa) from observations based on x axis (petal length) and y axis (petal width)



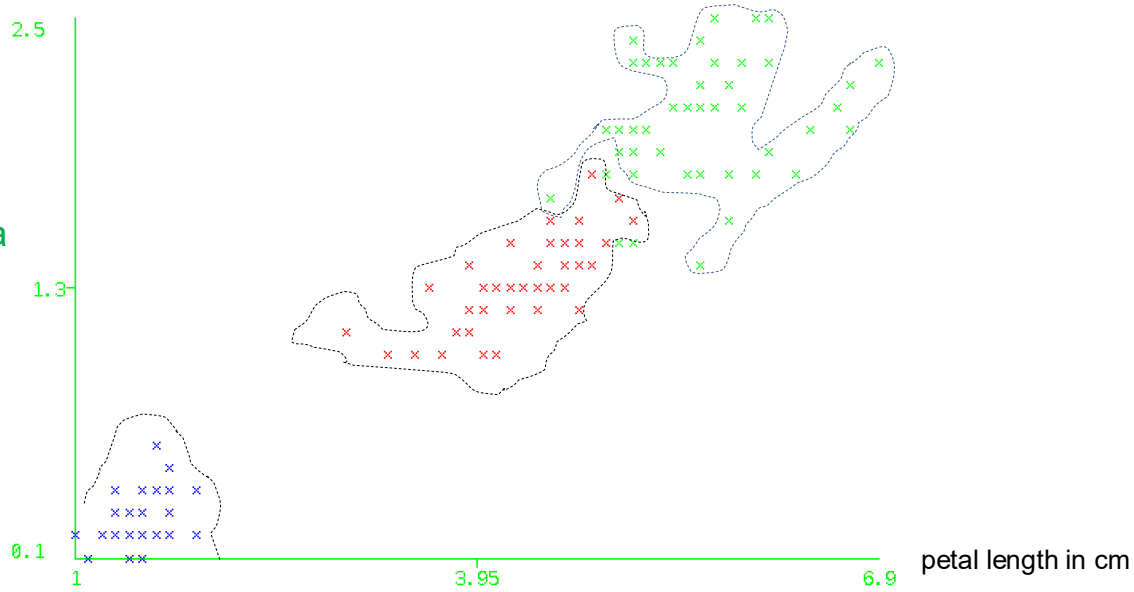



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# Supervised learning

- **Over-fitting**. A model learns the training data too closely and does not generalize. Poor performance on data other than the training dataset.

red: Iris versicolor
green: Iris virginica
blue: Iris setosa



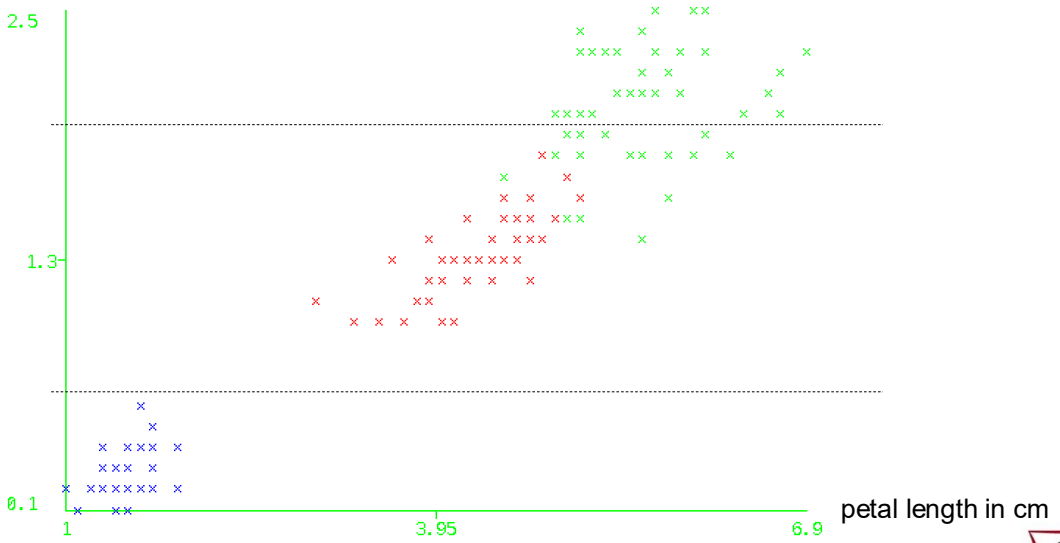



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# Supervised learning

- **Under-fitting**. A model does not fit the data well enough (little data for training, e. g. linear model with non-linear data, …). Poor performance on all data, including the training dataset.

red: Iris versicolor
green: Iris virginica
blue: Iris setosa

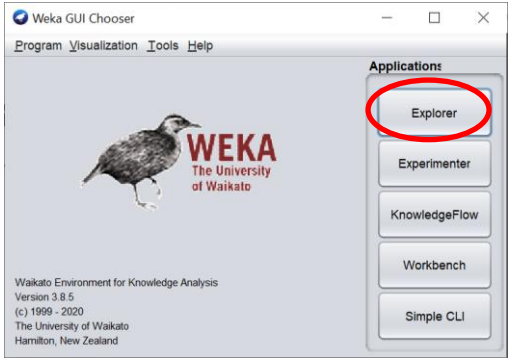


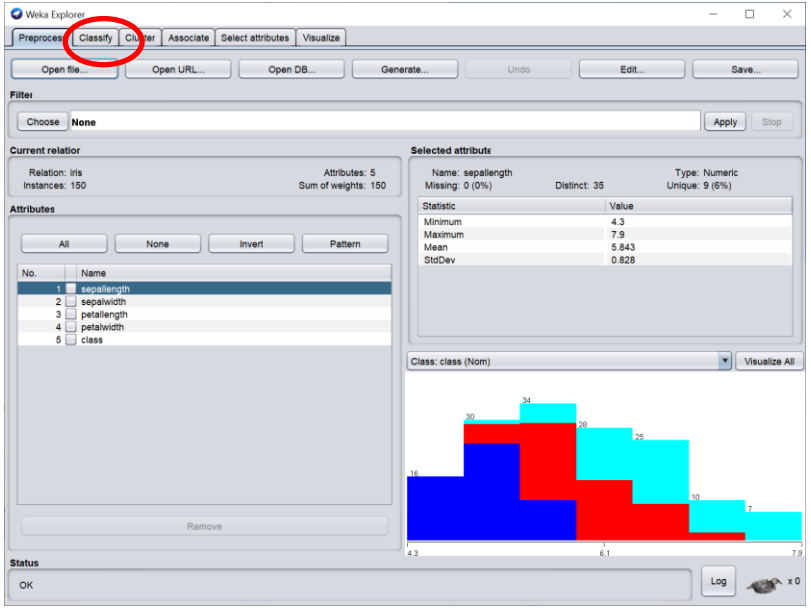



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# Classification with WEKA

- In the Explorer, open iris.arff, go to the “Classifiy” tab



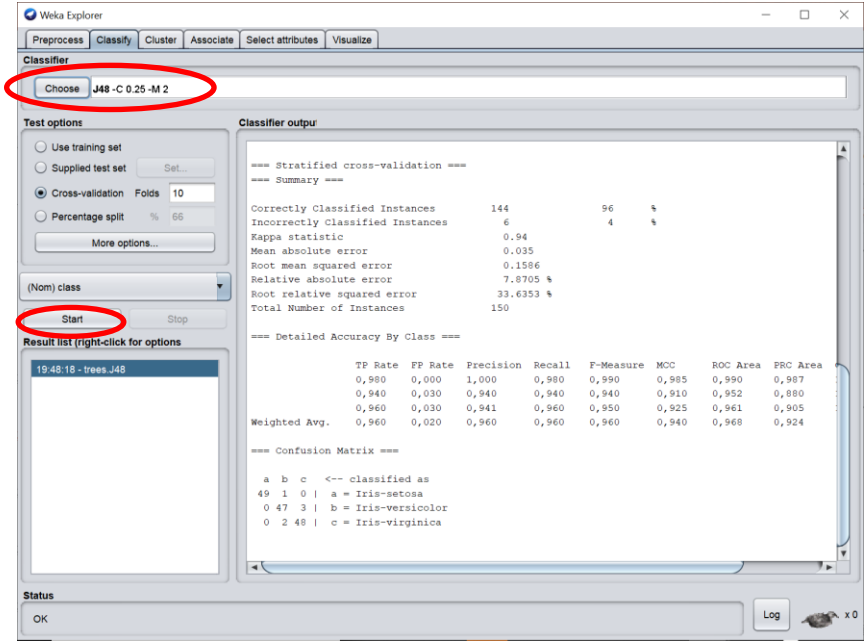






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# Classification with WEKA

- Select J48 (weka-classifier-trees), then click on Start

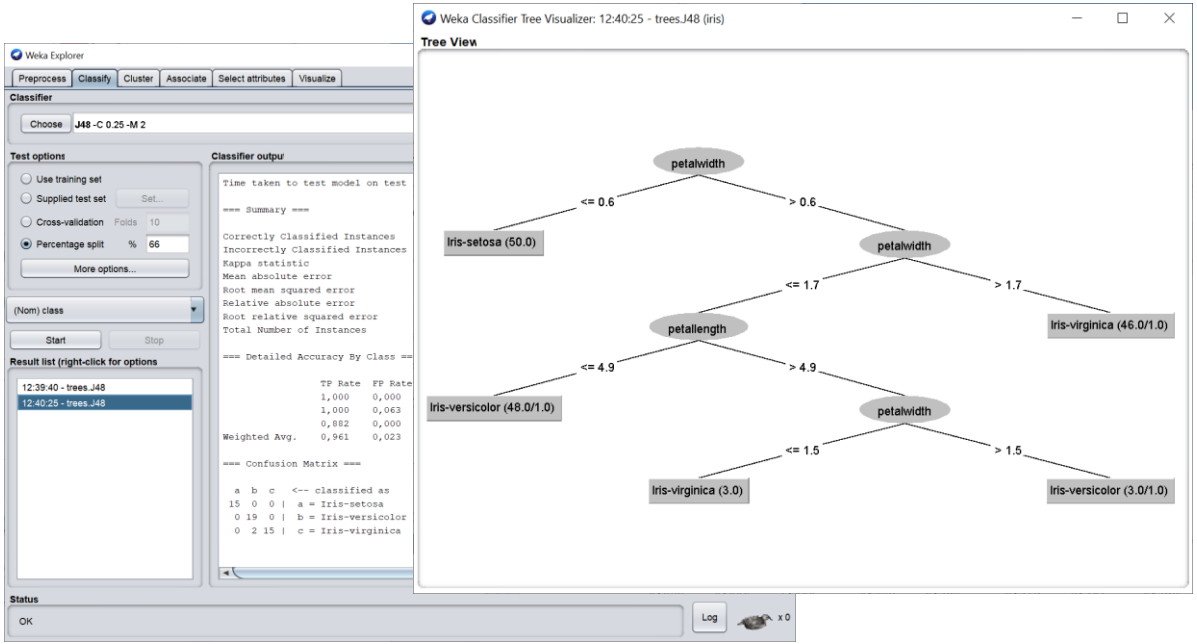





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# Classification with WEKA

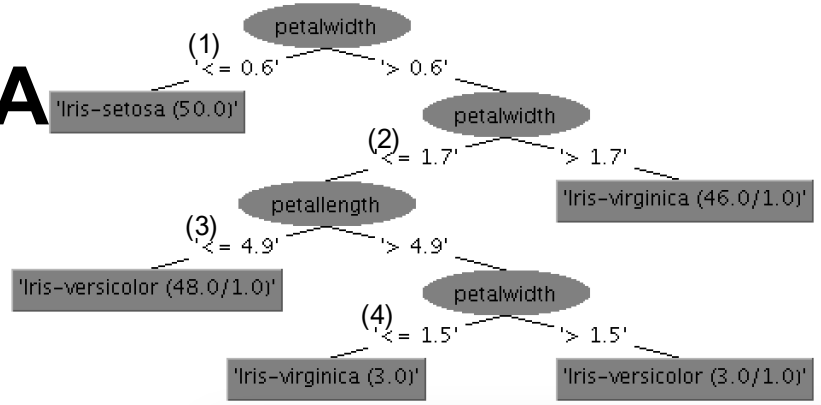
- Right-click on the result in “result list” and select “Visualize tree”

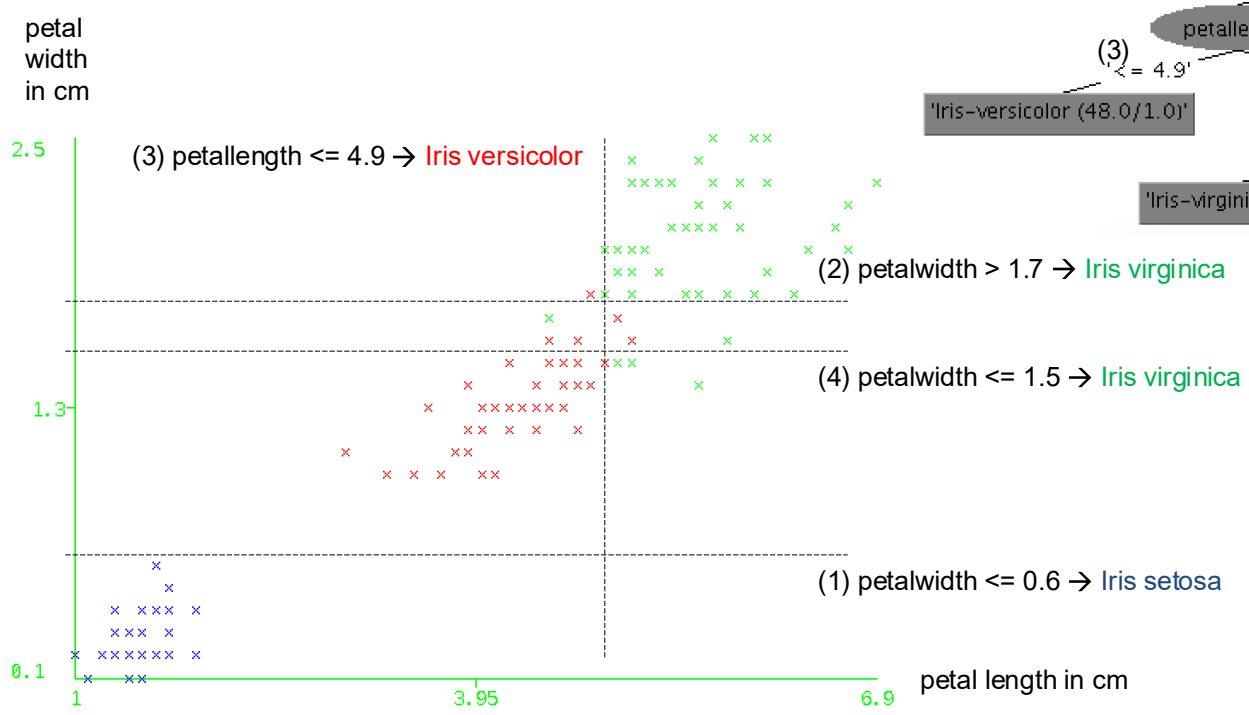





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# Classification with WEKA



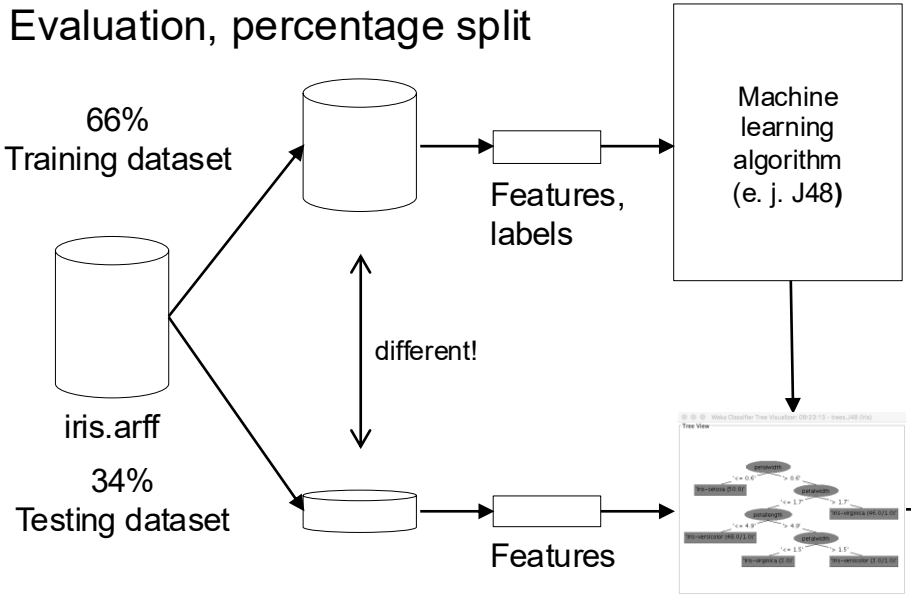



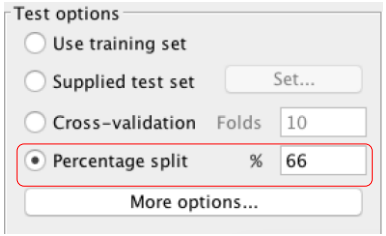



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# Classification with WEKA

- Evaluation, percentage split






```
Correctly Classified Instances          49               96.0784 %
Incorrectly Classified Instances         2                3.9216 %
Kappa statistic                          0.9408
Mean absolute error                      0.0396
Root mean squared error                  0.1579
Relative absolute error                  8.8979 %
Root relative squared error             33.4091 %
Total Number of Instances               51

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure
                   1         0          1          1         1
                   1         0.063      0.905      1         0.95
                   0.882     0          1          0.882     0.938
Weighted Avg.      0.961     0.023      0.965      0.961     0.961

=== Confusion Matrix ===

  a  b  c   <-- classified as
 15  0  0 |  a = Iris-setosa
  0 19  0 |  b = Iris-versicolor
  0  2 15 |  c = Iris-virginica
```



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# Classification with WEKA

- **Interpretation of evaluation results**, number of
  - **tp**: true positive
  - **tf**: true false
  - **fp**: false positive
  - **fn**: false negative

| | | True Value | |
|---|---|---|---|
| | | TRUE | FALSE |
| Classified As | TRUE | True Positive(TP) | False Positive(FP) |
| | FALSE | False Negative(FN) | True Negative(TN) |

- E.g., a COVID-19 test…
  - true positive: You have COVID-19 and you are tested positive.
  - true negative: You don't have COVID-19 and you are tested negative.
  - false positive: You don't have COVID-19 but you are tested positive.
  - false negative: You have COVID-19 but you are tested negative.



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# Classification with WEKA

- **Interpretation of evaluation results**, number of
  - **tp**: true positive
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| | | True Value | |
|---|---|---|---|
| | | TRUE | FALSE |
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| | FALSE | False Negative(FN) | True Negative(TN) |

- E.g., a COVID-19 test…
  - true positive: You have COVID-19 and you are tested positive.
  - true negative: You don't have COVID-19 and you are tested negative.
  - false positive: You don't have COVID-19 but you are tested positive.
  - false negative: You have COVID-19 but you are tested negative.



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# Classification with WEKA

- **Accuracy**
  - (tn + tp) / (tn + tp+ fp + fn)
  - Right results / all results

- E.g., a COVID-19 test…

| | | True Value | |
|---|---|---|---|
| | | TRUE | FALSE |
| Classified As | TRUE | 1 | 1 |
| | FALSE | 8 | 90 |

  - accuracy = (90+1)/(90+1+1+8) = 91%
  - only one true positive is detected in this test, 8 are not detected…

# Classification with WEKA

- **Precision**: tp / (tp + fp)
  - True Positive/Total Classified Positive
  - The ability of the classifier not to label as positive a sample that is negative;
  - Not many true values were predicted as false
  - Accuracy under the circumstance when a test result was positive
  - Useful when the **cost of fp is high** (e.g., a low precision COVID19 test would send many people to the hospital, even if they don't have the disease)

- Example
  - precision = 1 / (1 + 1) = 0.5
  - 50 % chance that you don't have COVID-19

# Classification with WEKA

- **Recall**: tp / (tp + fn)
  - Sensitivity, hit rate
  - the ability of the classifier to find all the positive samples; how many positive cases are identified out of the total positive cases;
  - important when the **cost of fn is high**. (e.g., cancer detection, a patient will miss a treatment)
  - The goal is to identify a bad thing (cancer)

- Example
  - recall = tp / (p + fn) = 1 / 9 = 0.11
  - Only 11% of the real positive cases are deteced



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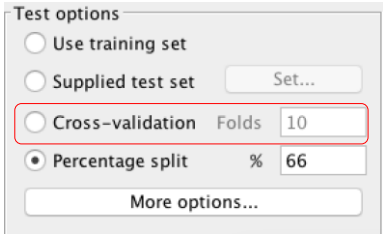
# Classification with WEKA

- Evaluation, percentage split
  - 150 instances, 99 used for training, 51 used for testing:
    - Correctly Classified Instances 49 96.0784 %
    - Incorrectly Classified Instances 2 3.9216 %

- Depending on the partitioning of the training and test data sets actual results may vary → in order to get a reliable performance estimate you need to average over several experimental runs → cross-validation

# Classification with WEKA

- Evaluation – 10-fold cross-validation
  - Randomly divide the data set into 10 parts (folds)
  - Use 9 folds for training (blue), 1 for testing (red)
  - Repeat 10 times
  - Average the results




- Stratified cross validation (default in Weka)
  - Ensure each fold has the right proportion of each class value

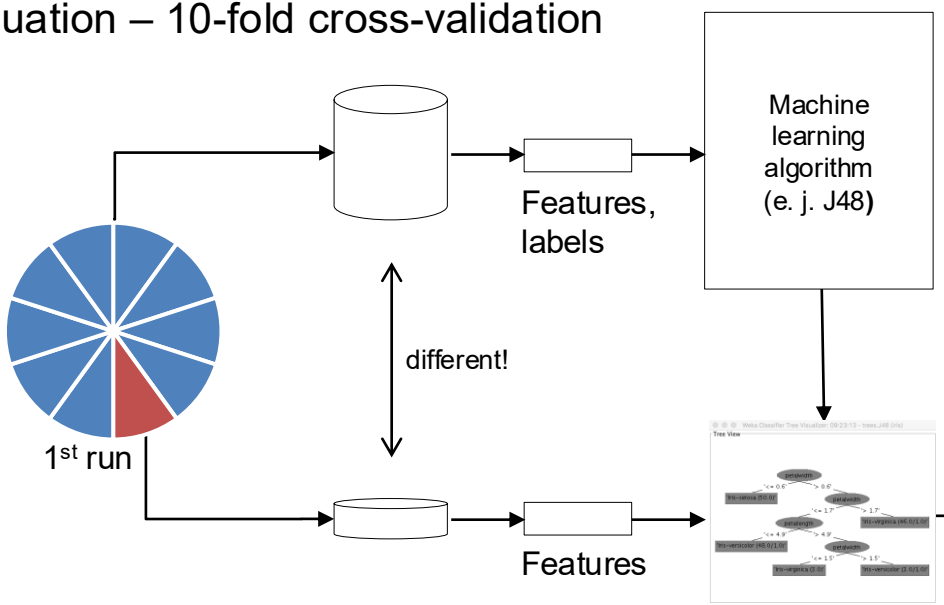
- Percentage split vs. cross evaluation
  - Lots of data → percentage split
  - Else stratified 10-fold cross-validation

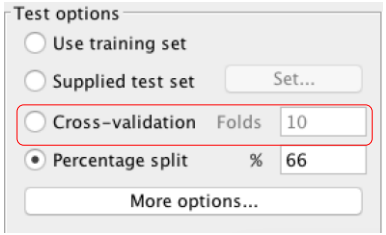


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# Classification with WEKA

- Evaluation – 10-fold cross-validation




```
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=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure
                   1         0          1         1         1
                   1         0.063      0.905     1         0.95
                   0.882     0          1         0.882     0.938
Weighted Avg.      0.961     0.023      0.965     0.961     0.961

=== Confusion Matrix ===

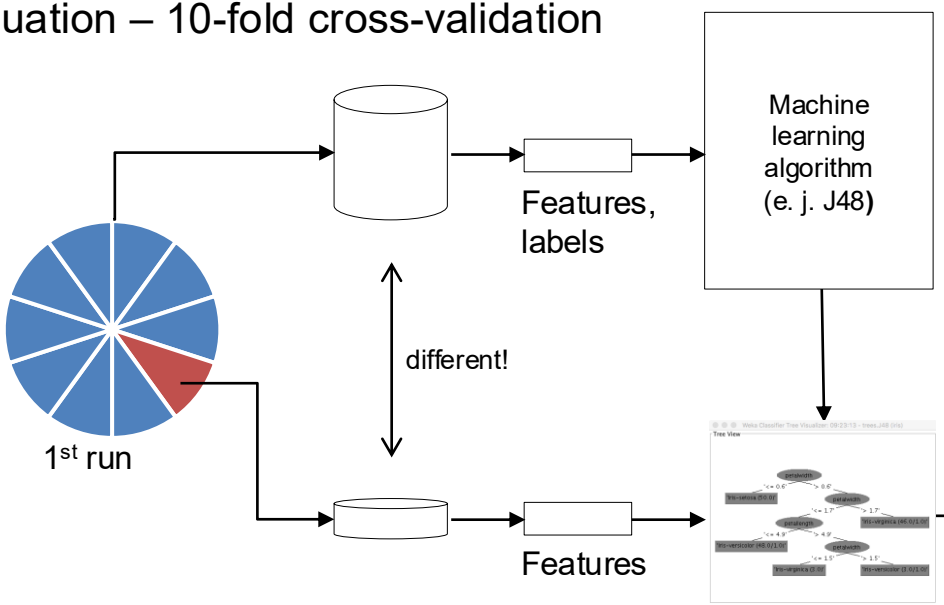
  a  b  c   <-- classified as
 15  0  0 |  a = Iris-setosa
  0 19  0 |  b = Iris-versicolor
  0  2 15 |  c = Iris-virginica
```

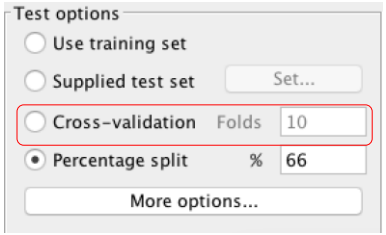


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# Classification with WEKA

- Evaluation – 10-fold cross-validation




```
Correctly Classified Instances          49               96.0784 %
Incorrectly Classified Instances         2                3.9216 %
Kappa statistic                          0.9408
Mean absolute error                      0.0396
Root mean squared error                  0.1579
Relative absolute error                  8.8979 %
Root relative squared error             33.4091 %
Total Number of Instances               51

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure
                   1         0          1         1         1
                   1         0.063      0.905     1         0.95
                   0.882     0          1         0.882     0.938
Weighted Avg.      0.961     0.023      0.965     0.961     0.961

=== Confusion Matrix ===

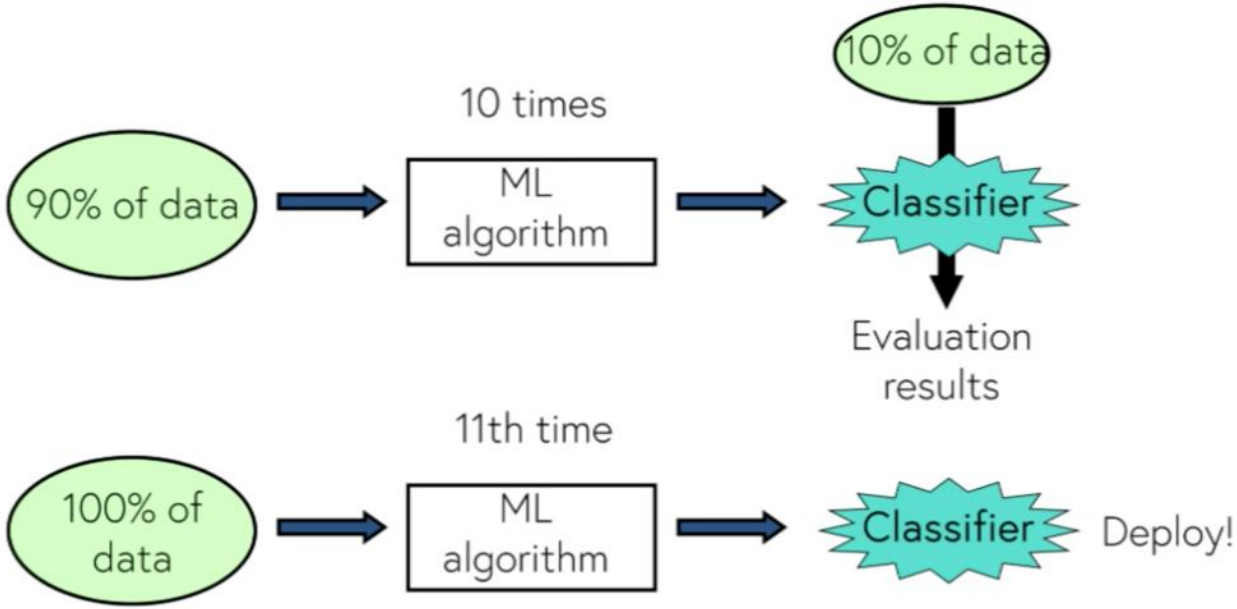
  a  b  c   <-- classified as
 15  0  0 |  a = Iris-setosa
  0 19  0 |  b = Iris-versicolor
  0  2 15 |  c = Iris-virginica
```



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# Classification with WEKA

- After cross-validation, Weka outputs an extra model built on the entire dataset






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# Classification with WEKA

- Confusion matrix
  - Each matrix element shows the number of test examples for which the actual class is the row, and the predicted class is the column.
  - Good results correspond to large numbers down the main diagonal and small, ideally zero, off-diagonal elements.

```
=== Confusion Matrix ===

  a  b  c   <-- classified as
 49  1  0 |  a = Iris-setosa
  0 47  3 |  b = Iris-versicolor
  0  2 48 |  c = Iris-virginica
```

  - 2 virginicas are mistakenly classified as versicolor
  - 1 setosas is mistakenly classified as versicolor
  - 3 versicolors  are mistakenly classified as virginica



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# Classification with scikit-learn

- We split the features into the attributes (sepal.length, …) and the class (variety)
- Next, we partition all our instances in a training and test set; by default, train_test_split() uses 75% for training and 25% for testing)

```
import pandas as pd
from sklearn.model_selection import train_test_split

iris = pd.read_csv('./iris.csv')

X = iris[['sepal.length', 'sepal.width', 'petal.length', 'petal.width']]
y = iris['variety']

X_train, X_test, y_train, y_test = train_test_split(X, y)
```



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# Classification with scikit-learn

- A classifier algorithm
  - is instantiated,
  - learns the training data (classifier.fit()),
  - and we get an evaluation with classfier.score() which reports an accuracy of 0.9473

```
classifier = tree.DecisionTreeClassifier()
classifier.fit(X_train, y_train)

print(classifier.score(X_test, y_test))
```



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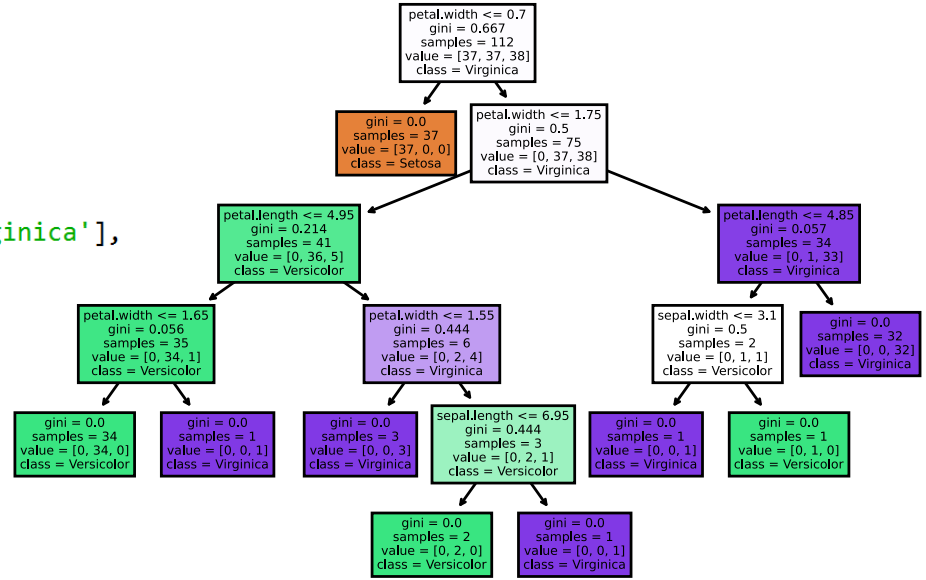
# Classification with scikit-learn

- With a decision tree, we can plot the tree with tree.plot_tree(classifier)

```
classifier = tree.DecisionTreeClassifier()
classifier.fit(X_train, y_train)

print(classifier.score(X_test, y_test))

tree.plot_tree(classifier, feature_names=X.columns,
                  class_names=['Setosa', 'Versicolor', 'Virginica'],
                  filled=True)
```






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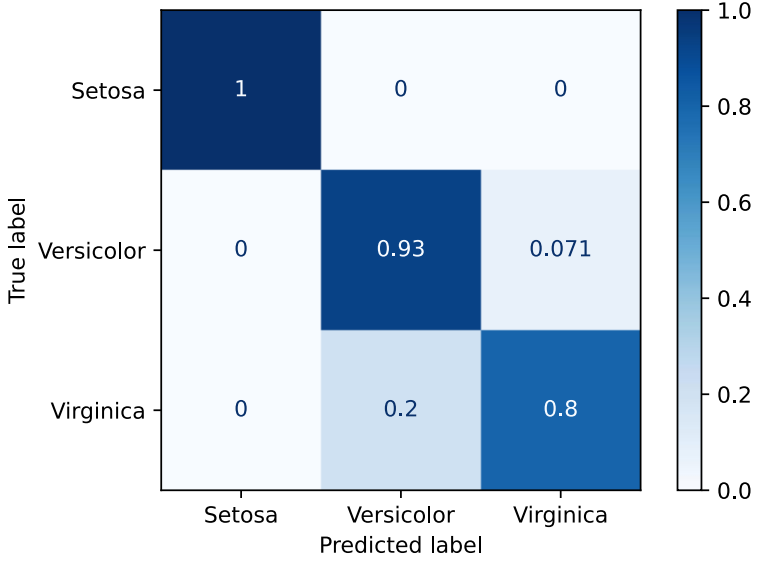
# Classification with scikit-learn

- Confusion matrix

```
classifier = tree.DecisionTreeClassifier()
classifier.fit(X_train, y_train)

disp = ConfusionMatrixDisplay.from_estimator(
        classifier,
        X_test,
        y_test,
        display_labels=['Setosa', 'Versicolor', 'Virginica'],
        cmap=plt.cm.Blues,
        normalize='true'
    )

print(disp.confusion_matrix)
plt.show()
```






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# Classification with scikit-learn

- classification_report()

```
classifier = tree.DecisionTreeClassifier()
classifier.fit(X_train, y_train)

y_pred = classifier.predict(X_test)
print(classification_report(y_test, y_pred))
```

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Setosa | 1.00 | 1.00 | 1.00 | 14 |
| Versicolor | 0.92 | 0.92 | 0.92 | 12 |
| Virginica | 0.92 | 0.92 | 0.92 | 12 |
| accuracy | | | 0.95 | 38 |
| macro avg | 0.94 | 0.94 | 0.94 | 38 |
| weighted avg | 0.95 | 0.95 | 0.95 | 38 |



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# Classification with scikit-learn

- Code generation from scikit-learn models
  - m2cgen
    - https://github.com/BayesWitnesses/m2cgen
    - Target languages: Python, C, Java, Go, JavaScript, Visual Basic, C#, PowerShell, R, PHP, Dart, Haskell, Ruby, F#, Rust, Elixir
  - sklearn-porter
    - https://github.com/nok/sklearn-porter
    - Target languages: C, Go, Java, JS, PHP, Ruby
  - emlearn
    - https://github.com/emlearn/emlearn
    - Only C-code

# Classification with scikit-learn

- Example with m2cgen

```
from sklearn import tree
import pandas as pd
from sklearn.model_selection import train_test_split

import m2cgen as m2c

iris = pd.read_csv('./iris.csv')

X = iris[['sepal.length', 'sepal.width', 'petal.length', 'petal.width']]
y = iris['variety']

X_train, X_test, y_train, y_test = train_test_split(X, y)

classifier = tree.DecisionTreeClassifier()
classifier.fit(X_train, y_train)

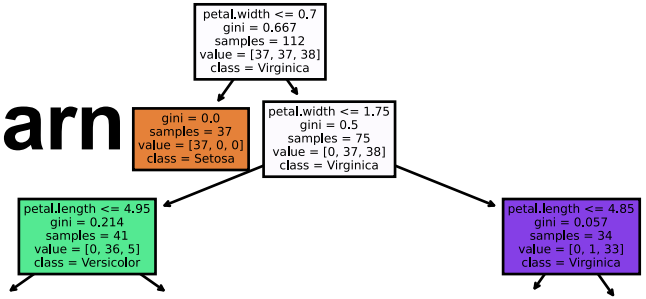
code = m2c.export_to_c(classifier)

f = open( 'iris.c', 'w' )
f.write(code)
f.close()
```



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# Classification with scikit-learn




- Example with m2cgen

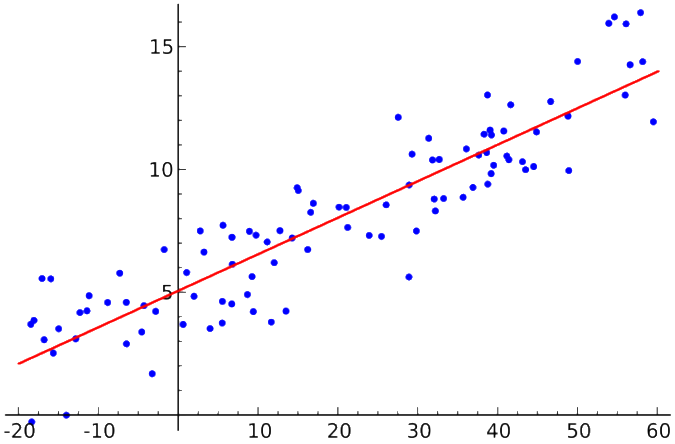
```
#include <string.h>
void score(double * input, double * output) {
    double var0[3];
    if ((input[3]) <= (0.7000000029802322)) {
        memcpy(var0, (double[]){1.0, 0.0, 0.0}, 3 * sizeof(double));
    } else {
        if ((input[3]) <= (1.75)) {
            if ((input[2]) <= (4.950000047683716)) {
                if ((input[3]) <= (1.6500000357627869)) {
                    memcpy(var0, (double[]){0.0, 1.0, 0.0}, 3 * sizeof(double));
                } else {
                    memcpy(var0, (double[]){0.0, 0.0, 1.0}, 3 * sizeof(double));
                }
            } else {
```



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# Regression

- Predict **values** from observations






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# Regression

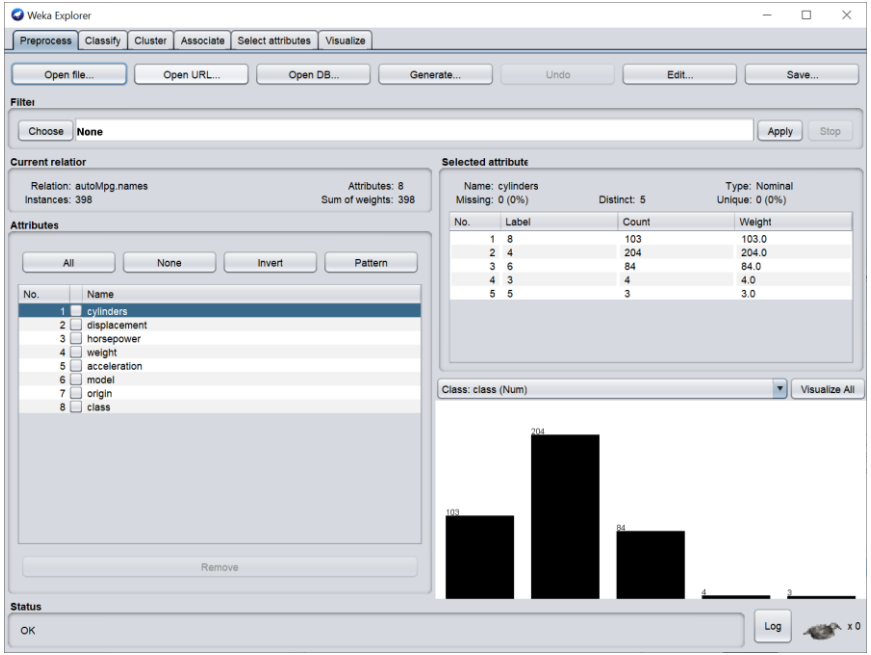
- Example: the task is to predict the miles per gallon, based on
  - Cylinders (8, 4, 6, 3, 5)
  - Displacement
  - Horsepower
  - Weight
  - Acceleration
  - Model year (70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82)
  - Origin (1, 3, 2), where 1=US, 2=Europe, 3=Japan
- Data is available as autompg.arff (see Moodle)



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# Regression

- Linear regression in Weka

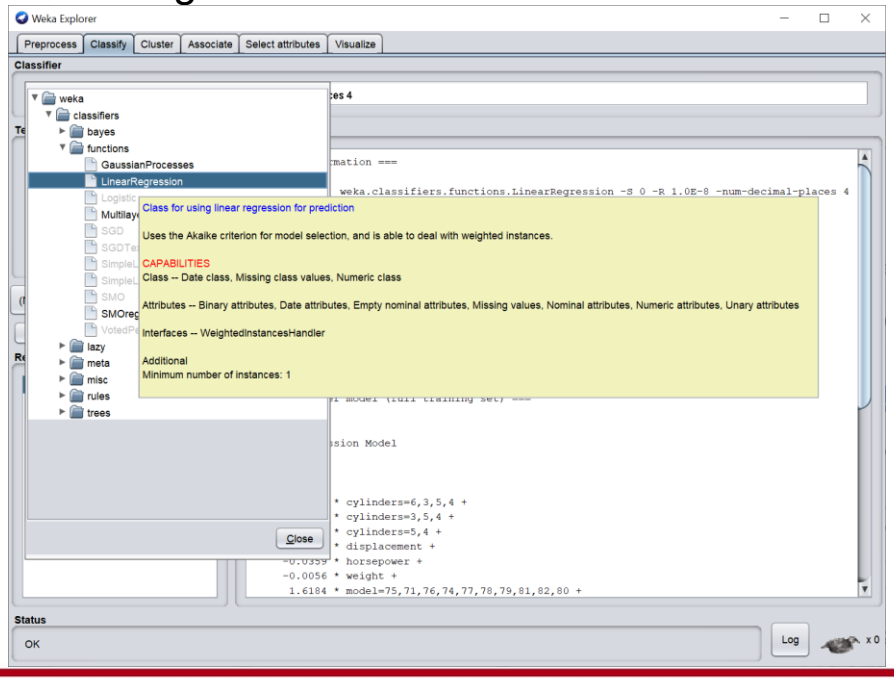





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# Regression

- Linear regression in Weka

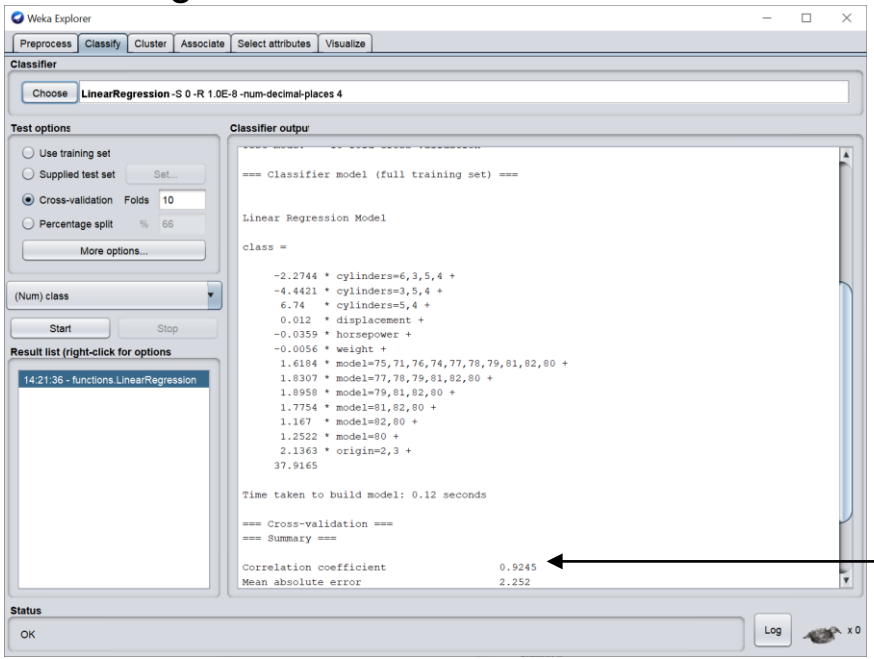





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# Regression

- Linear regression in Weka



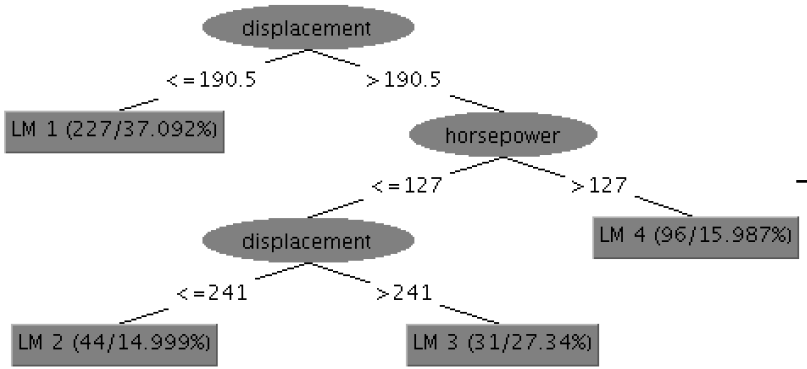

Correlation coefficient: 0.92



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# Regression

- Non-linear regression
  - Practical problems often require non-linear models
  - Model tree
    - Each leaf has a linear regression model
    - Linear patches approximate continuous function



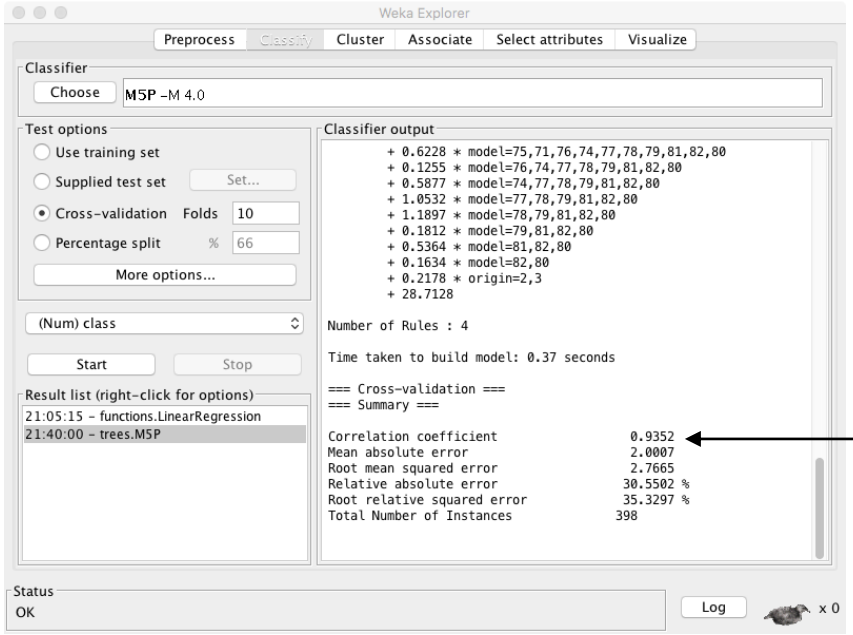

```
LM num: 4
class =
        0.0024 * displacement
        - 0.0278 * horsepower
        - 0.0016 * weight
        - 0.3653 * acceleration
        + 0.6228 * model=75,71,76,74,77,78,79,81,82,80
        + 0.1255 * model=76,74,77,78,79,81,82,80
        + 0.5877 * model=74,77,78,79,81,82,80
        + 1.0532 * model=77,78,79,81,82,80
        + 1.1897 * model=78,79,81,82,80
        + 0.1812 * model=79,81,82,80
        + 0.5364 * model=81,82,80
        + 0.1634 * model=82,80
        + 0.2178 * origin=2,3
        + 28.7128
```



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# Regression

- Non-linear regression in Weka, “M5P”



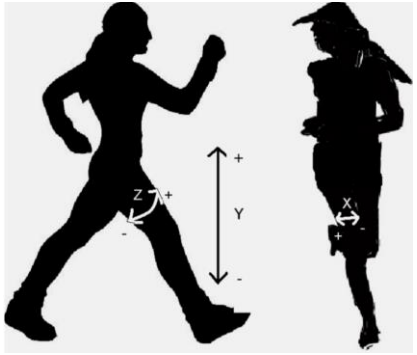

Correlation coefficient: 0.93



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# Activity Recognition

- J. R. Kwapisz, G. M. Weiss, S. A. Moore. Activity recognition using cell phone accelerometers. ACM SIGKDD Explorations Newsletter Volume 12 Issue 2 December 2010
  - 29 subjects carried an Android phone in their front pants leg pocket and were asked to walk, jog, ascend stairs, descend stairs, sit, and stand for specific periods of time
  - sampling frequency of 20 Hz






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# Activity Recognition

- Data collection (file WISDM_ar_v1.1_raw.txt, 50MB, 1098204 lines)
  - Subject
  - activity (walking, jogging, ascending stairs, descending stairs, sitting, standing) = label
  - timestamp
  - x, y, and z axes in m/s2

```
33,Jogging,49105962326000,-0.6946377,12.680544,0.50395286;
33,Jogging,49106062271000,5.012288,11.264028,0.95342433;
33,Jogging,49106112167000,4.903325,10.882658,-0.08172209;
33,Jogging,49106222305000,-0.61291564,18.496431,3.0237172;
33,Jogging,49106332290000,-1.1849703,12.108489,7.205164;
```



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# Activity Recognition

- Preprocessing and feature extraction
  - Standard classification algorithms cannot be directly applied to raw time-series accelerometer data.
  - Instead, raw time series data is transformed into so called examples.
  - Data is divided into **10-second segments** and then features that are based on the 200 readings are calculated.
  - A 10-second duration provides **sufficient time to capture several repetitions of the (repetitive) motions** involved in some of the six activities.
  - Compared to a 20-second duration, the 10 second duration yielded slightly better results (as well as twice as many training examples).

# Activity Recognition

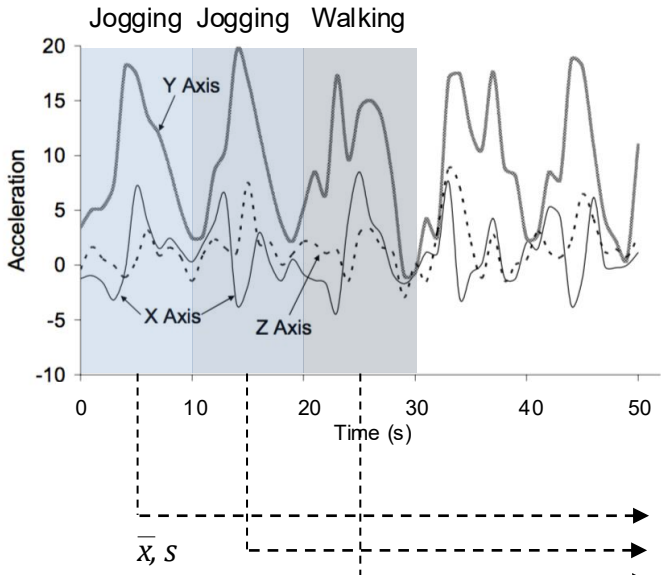
- Preprocessing and feature extraction
  - Some features that are typically used in activity recognition are
  - Average: $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
  - Standard deviation: $s = \sqrt{\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$
  - Time between peaks: The repetitive activities, like walking, tend to generate repeating waves for each axis and this feature tries to measure the time between successive peaks
  - Spectrum



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# Activity Recognition

- Preprocessing and feature extraction



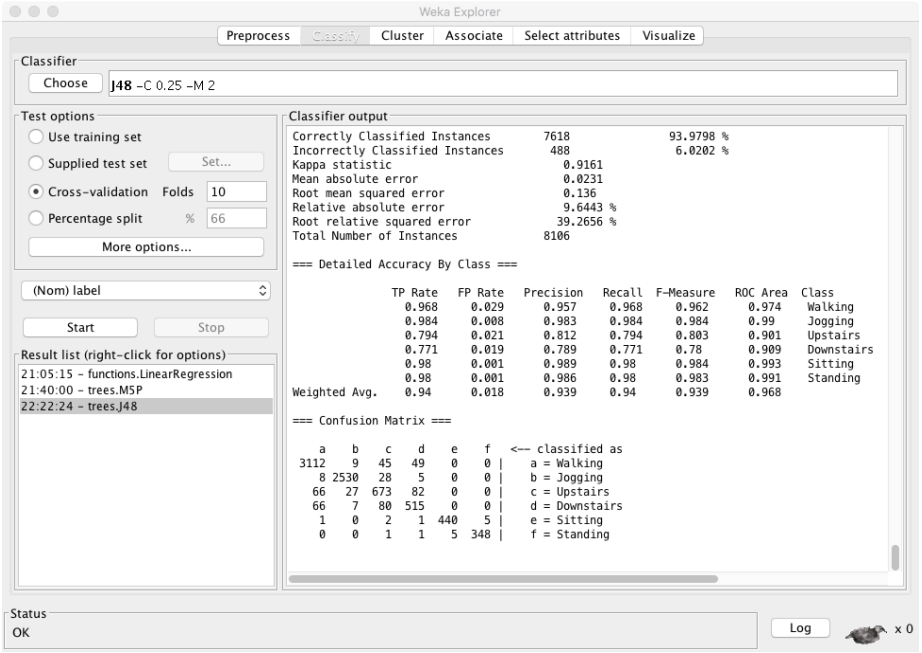

```
@relation activity_recognition
@attribute "subject" numeric
@attribute "xmean" numeric
@attribute "ymean" numeric
@attribute "zmean" numeric
@attribute "xdev" numeric
@attribute "ydev" numeric
@attribute "zdev" numeric
@attribute label {Walking, Jogging, Upstairs, Downstairs, Sitting, Standing}
@data
33 -0.310289 8.015894 2.055396 3.792297 7.914810 4.937366 Jogging
33 -0.535024 8.842585 1.225406 4.650245 8.562298 5.609649 Jogging
33 -0.236654 9.320893 0.390947 3.004852 5.074578 3.929691 Walking
33 -0.728263 9.700667 0.439575 2.715680 4.697020 3.866445 Walking
```



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# Activity Recognition

- Classification with J48






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# Assignment 2

- Use Weka, scikit-learn, or any other machine learning tool of your choice to generate a decision tree which classifies activities (WISDM_ar_v1.1_raw.txt).
  - For the feature extraction, use an example duration of approx. 10 seconds and extract the average and standard deviation for each axis.
  - Don’t forget to remove the first column (subject id) before training the classifier.
  - Evaluate the classifier’s performance in your machine learning environement and then implement the decision tree in an app on a mobile platform of your choice (as discussed last week and in assignment 1).



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